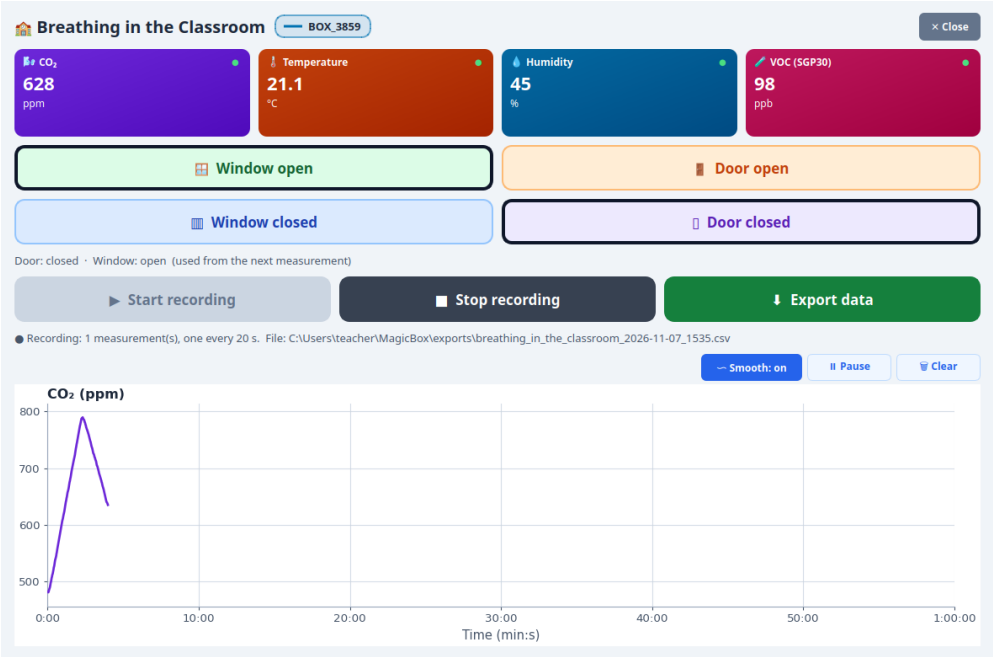
Breathing in the Classroom

A whole lesson recorded every 20 seconds, with the door and the window marked on the timeline

Level: Explorer and Scientist — ages 11–19 (Romanian school years V–XII) • **Time in class:** one 50-minute lesson of recording; a second lesson for the analysis in Excel • **Sensors:** SCD41 (CO_2); optional SGP30 or SGP41 (VOC) and BME280 (temperature, humidity)



The measuring page during a recording. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Explorer and Scientist — ages 11–19 (Romanian school years V–XII). In years V–VI pupils read the shape of the curve and link each bend to a door or a window; in years VII–VIII they explain the rise through respiration (biology) and the water vapour we breathe out (physics); at upper secondary level they work out the rate of rise and the decay after airing in a spreadsheet, up to a logarithmic fit.

Assumed knowledge

Reading a graph over time. Percentages. For the Scientist level, building a chart in Excel or LibreOffice Calc and the slope of a straight line; the natural logarithm for the extension.

Time in class

The box records a whole lesson (45–50 min) while the class works on something else; ten minutes of it go to the prediction and to the airing. The analysis takes a second lesson, at the computer.

Teacher preparation

15 min — switch the box on at least ten minutes beforehand, better still in the morning: the SGP41's VOC index learns the room's air over its first hours. Open the activity and look at the start page: **each sensor box should be green**. Put the box at the head height of a seated pupil, at least a metre from anyone, **away from the window, the door and the radiator**; next to an open window it measures the draught, not the room. Decide in advance how you will air: the window, the door, or both.

Sensor settling time — CO_2: 30 seconds after power-up, then a new value every five seconds. VOC: 15 seconds (SGP30) or about a minute (SGP41). Temperature and humidity: at once. The first row is written as soon as you press **Start recording**, so wait a minute after switching the box on; a value older than ten seconds leaves its cell empty rather than repeating an old number.

Sensors and boxes — The **SCD41** (CO_2) is the one that matters. The **BME280** adds temperature and humidity; without it, both come from the SCD41, which warms itself and reads them less accurately (the app says so on both pages). The **SGP30** (TVOC, in ppb) or the **SGP41** (VOC index) fills the VOC column; these two need firmware 2.5.0 or newer. A usual classroom box with an SCD41 and a BME280 runs the lesson as it is, with the VOC column empty.

Three routes

Short — One 50-minute lesson: the prediction and the start of the recording (5), ordinary teaching with the room closed (20–25), the airing (5–10), the room closed again until the bell (10), **Stop recording** and the export (2). The analysis follows in lesson 2.

Full — Short version, 25 min: record from the start of a written test, open the window for five minutes at the end and look at the curve on the screen with the class, without Excel.

Project — Three-lesson project: the same class, the same room, three airings — the door alone, the window alone, the window and the door together — and one chart with the three decay curves side by side.

The application starts in English. Buttons are named here exactly as they appear on the screen.

Where to find the screen

From version 5.9 the lesson appears on the **Activities** page by itself, once. If someone has hidden it, open **Settings**, go to **Visible activities** and tick *Breathing in the Classroom* again. The tick stays on that computer.

1. Where this fits in the curriculum

- **Biology** — in year VII, human respiration: exhaled air carries about a hundred times more CO_2 than the air outside, and the room's curve is the class breathing.
- **Physics** — in year VIII, heat and water vapour: thirty seated pupils give off about as much heat as an electric heater, and the humidity climbs with every breath.
- **Geography** — in year V, the composition of the atmosphere; in year IX, air quality and ventilation of buildings.
- **Mathematics** — in years VI–VII, rate of change from a table; in years X–XI, the exponential function and the logarithm: the CO_2 in excess of the outdoor air falls exponentially once the window is open.

The core skill is **reading a long record against a log of events**. Pupils see that a curve is a story of what happened in the room, and that each bend has a cause they can name.

2. Time plan

Route	Time	What happens
Lesson 1	**50 min**	The prediction and **Start recording** (5), the room closed (20–25), the airing with its button pressed (5–10), the room closed again (10), **Stop recording** and **Export data** (2). The teaching itself goes on all the time.
Short	25 min	A written test with the room closed, five minutes of open window at the end, the curve read on the screen.
Between lessons	—	The file `breathing_in_the_classroom_...csv` is copied to the computers the pupils will use.
Lesson 2	50 min	In Excel: the chart of CO_2 against time, the rate of rise, the time it took the excess to halve after the window opened.

3. Materials

Item	Per class	Notes
MagicBox with an SCD41, preferably also a BME280 and an SGP30 or SGP41	1	One box is enough: the room has a single air. A second box in another room gives a comparison.
Laptop with the app	1	Over USB or WiFi; it stays on for the whole lesson.
A room with a window and a door that open	1	The door to a corridor counts; a tilted window counts too, and makes a useful comparison.
The pupil sheet	1 per pupil	Its event log is filled in by the pupil who presses the buttons.
Excel or LibreOffice Calc	1 per group	For lesson 2.

Preparation the day before. Record ten minutes in the empty room and press each of the four buttons once. Then open the saved file in Excel: you should see one row every 20 seconds, the CO_2 near 420–500 ppm in an aired room, and the door and window columns filled from the moment you pressed them. Ten minutes now save the lesson from a box that was never checked.

4. Scientific background for the teacher

What a class adds to the air of a closed room

Carbon dioxide. Outdoor air holds about 420 ppm of CO_2 (the global mean measured by NOAA was about 423 ppm in 2024); the air we breathe out holds about 4 %, some 40 000 ppm. A seated pupil breathes out roughly 0.2 L of CO_2 a minute, so thirty pupils in a 170 m^3 room with no ventilation at all would add about 35 ppm a minute. Real rooms leak, and the measured rise is usually slower. At classroom levels CO_2 is not a poison; it is a **tracer of exhaled air**, showing how much of what we breathe has already passed through somebody's lungs.

The fall after opening. Fresh air replaces room air at some rate, measured in **air changes per hour** (n). The CO_2 *in excess* of the outdoor level then falls by the same fraction every minute: it decays exponentially, $C(t) - C_{\text{out}} = (C_0 - C_{\text{out}})\, e^{-nt}$, and halves in $t_{1/2} = 0.69/n$. While the class stays in the room the breathing goes on, so the CO_2 settles at a level above the outdoor one instead of reaching it. As an order of magnitude only: a closed room changes

its air less than once an hour, a tilted window or an open door to a corridor a few times, a wide-open window with the door open across the room ten times or more. Your room will have its own numbers; measuring them is the point.

Water vapour and heat. A seated person breathes out some 40–50 g of water an hour and gives off about 100 W of heat. Thirty pupils add more than a litre of water an hour and as much heat as a small electric heater, so the humidity and the temperature both climb during the lesson. In cold weather, airing lowers the relative humidity: outdoor air holds little water, and once warmed in the room its relative humidity is low.

VOC. Volatile organic compounds come from markers, glue, perfume and cleaning products, and from people too (breath and skin). The SGP30 gives an estimate of their total, **TVOC**, in ppb. The SGP41 gives a **VOC index** from 1 to 500 in which 100 is the room's average over the last hours. Both are qualitative indicators: they show *that* something changed, not which gas, and not how much of it.

4.1 Reference values and what they mean

Quantity	Reference	What it means
CO_2 outdoors	about 420 ppm	The floor: airing cannot bring the room below it.
CO_2 indoors	about 1000 ppm	The usual classroom guidance (the German Federal Environment Agency, UBA; EN 16798-1 sets classes relative to the outdoor air). Above 2000 ppm the UBA calls the air unacceptable; closed classrooms often get there within a lesson.
Relative humidity	about 40–60 %	A comfortable range indoors, not a health limit.
Temperature	about 20–22 °C	Comfort in a classroom in winter.
TVOC (SGP30)	no official threshold	Compare the parts of the lesson with one another, not with a number from elsewhere.
VOC index (SGP41)	100 = the room's usual air	A comparison with the same room's recent past, not a concentration.

Studies of schools find poorer results on attention tests in classrooms with little ventilation. Whether the CO_2 itself is responsible, or the other things that build up with it, is still debated; present 1000 ppm to pupils as a sign that it is time to air, not as a danger line.

5. Lesson flow

The screen opens on a **start page** with the three sensor boxes the activity needs — **CO_2 (SCD41)**, **VOC (SGP30 or SGP41)**, **Temperature and humidity (BME280)** — each green with **The box sees it** or red with **Not seen by the box**, and the button **Start**. Missing sensors do not stop the lesson: their columns stay empty. The **measuring page** holds the four cards, the four marking buttons, the recording buttons and the CO_2 chart.

1. **The prediction (3 min).** On the pupil sheet each pupil sketches the CO_2 curve for the lesson: where it starts, how high it gets, what happens when the window opens. Many pupils draw a fall straight back to the starting value.
2. **The start (2 min).** Air the room during the break, then close the door and the window. On the measuring page press **Window closed** and **Door closed**, then **Start recording**. One pupil is the recorder: they press the buttons and write the time of every event in the log on the sheet.

Technical note — what the screen shows and what is saved

The cards. CO_2 in ppm, temperature in °C, humidity in %, and VOC: the card reads **VOC (SGP30)** in ppb, or **VOC index (SGP41)** without a unit when the box carries an SGP41. Without a BME280 a line in orange says that temperature and humidity come from the SCD41.
The marks. Window open, **Door open**, **Window closed**, **Door closed**: the open and closed buttons of the same opening exclude each other. A line under the buttons repeats the current state; a mark counts **from the next row**, so it reaches the file within 20 seconds. Before the first mark the door and window cells are empty.
The recording. Start recording writes the first row at once, then one every 20 seconds, about 150 for a lesson; **Stop recording** ends it. The status line counts the rows and gives the file's path. The chart shows the last hour of CO_2.
The file that is always there. Every row is saved at once to a CSV in the folder `MagicBox\exports`, named `breathing_in_the_classroom_<YYYY-MM-DD_HHMM>.csv`. A closed window or a flat laptop battery loses nothing already recorded.
Export data. The button saves a copy as an Excel workbook (`.xlsx`, sheets `Data` and `Legend`) or as a CSV. The column names are always in English, whatever the language of the app: `Timestamp`, `Elapsed (s)`, `CO2 (ppm)`, `Temperature (°C)`, `Humidity (%)`, `VOC`, `Door open (1 = yes)`, `Window open (1 = yes)`. The `VOC` column holds ppb from an SGP30 or the index from an SGP41; the `Legend` sheet says which. In the door and window columns 1 means open, 0 closed, and an empty cell "not marked yet". A CSV saved with the app in English uses "," between columns and a decimal point; in Romanian, French and German it uses ";" and a decimal comma, so Excel set to that language opens it directly.

3. **The room closed (20–25 min).** The lesson goes on as usual. Glance at the chart now and then: the CO_2 climbs almost in a straight line, faster in a small room or a full one. Nobody breathes over the box.

4. **The airing (5–10 min).** At an agreed moment open the window wide (and, in the project route, the door) and the recorder presses **Window open**. The curve turns down within a minute. Close it again after five to ten minutes and press **Window closed**.

5. **To the bell (10 min).** The room is closed again and the curve climbs again, from a lower start. Before the bell press **Stop recording**, then **Export data** and choose Excel.

6. **Lesson 2: the analysis in Excel (50 min).** A line chart of `CO2 (ppm)` against `Elapsed (s)`; the door and window columns beside it show where each event falls. Pupils compute the rate of rise in ppm per minute over the closed part and read off how long the excess above 420 ppm took to halve once the window opened. Scientist level: plot $\ln(CO_2 - 420)$ against time during the airing; the slope of the straight part is $-n$, the air changes per second (multiply by 3600 for the hour).

The investigation, in one table

The question. How fast does a class fill its room with exhaled air, and how fast does opening the window or the door empty it again?
The hypothesis. Sketched before measuring: the starting value, the highest value, and the shape of the fall after the window opens.
What I change. The state of the window and of the door, at moments written in the log.
What I measure. CO_2, temperature, humidity and VOC, one row every 20 seconds, with the door and the window marked.
What I keep the same. Where the box stands, the number of people in the room, nobody within a metre of the box.
How many repeats. One lesson for each kind of airing (door, window, both), in the same

room and with the same class; the decay curves are then comparable.
What makes a reading valid — and what makes it an artefact. A pupil leaning over the box gives a CO_2 spike of several hundred ppm that lasts less than a minute; a box beside the open window drops to 420 ppm at once while the rest of the room is still stale; a mark pressed five minutes late moves the event in the file.

6. Analysis after the activity

6.1 What the data should look like

Quantity	What you usually see	Why
CO_2	a nearly straight rise with the room closed; a curved fall after opening that slows down near the bottom	The source is the class; the fall is exponential towards a level above 420 ppm.
Humidity	a rise of a few per cent over the lesson; in winter a drop while the window is open	Breath carries water; cold outdoor air holds little.
Temperature	a rise of a degree or two; a dip while airing and a quick recovery	Thirty bodies heat the room; walls and furniture keep their heat.
VOC	small steps when markers or perfume are used; usually a fall on airing	Sources mostly inside; a qualitative signal.

An illustrative example, not a measurement: the CO_2 goes from 650 to 1450 ppm in the first 25 minutes, a rise of 32 ppm a minute. The window opens; the excess above 420 ppm, 1030 ppm, falls to 515 ppm in four minutes, so $t_{1/2} = 4$ min and $n \approx 0.69/4 \approx 0.17$ per minute, about ten air changes an hour. **Do not look for a "correct" number**: rooms, classes and weather differ. Look at the shapes and at how the door compares with the window.

6.2 Questions to ask afterwards

1. How many ppm did the CO_2 gain per minute with the room closed? How long until it passed 1000 ppm?
2. Why does the curve fall quickly at first and more slowly later?
3. Why can the room never go below about 420 ppm, however long the window stays open?
4. Which emptied the room faster, the door or the window? What was on the other side of the door?
5. The humidity fell while the window was open, although it was raining outside. How can that be?

6.3 Assessment ideas

- **Low stakes:** the pupil labels each bend of the printed curve with the event from the log.
- **Standard:** the pupil computes the rate of rise and the time for the excess to halve, and explains both in a sentence.
- **Extension:** the pupil estimates the air changes per hour from a logarithmic fit and proposes how often this room should be aired, with a stated limitation.

Common pitfalls and misconceptions

"The window brings the air back to how it was." Only while the room is empty. With the class inside, the CO_2 settles at a level above the outdoor one, and closing the window starts the climb again.
Marks pressed late or forgotten. A mark counts from the next row. If the recorder forgets, write the real time in the log and correct the cell in Excel; do not guess from the curve, or the analysis becomes circular.
The box by the window. It measures the incoming air, not the room. Keep it in the middle of the room, at a pupil's head height.
Temperature from the SCD41. Without a BME280 the temperature reads a few degrees high, because the CO_2 sensor warms itself. Changes over the lesson are still meaningful; the absolute value is not.
The VOC index read as a concentration. An index of 150 means "more than usual in this room", not "one and a half times the limit".
A CSV opened in another language. A CSV saved with the app in English and opened by a Romanian Excel lands in one column. Use the `.xlsx` export, or save the CSV with the app in the language of the spreadsheet.

Safety and classroom management

Windows are opened by the teacher; nobody climbs on a sill, a radiator or a chair.
Cold. Air briefly and fully; move pupils away from the window while it is open.
Doors. A draught can slam a door: wedge it open, and keep fingers away from the hinge side.
No sprays. If you want to show the VOC rising, an open marker next to the box is enough; some pupils have asthma.

Adaptations for pupils with special educational needs

The recorder's role. Pressing a button and writing a time suits a pupil who finds it hard to sit through the lesson; the job has a clear beginning and end.
One quantity per pupil. In lesson 2, pupils who tire of tables follow only the CO_2 and say for each part of the lesson "up", "down" or "the same".
Reading the chart. Print the chart large, with the events marked by hand; a finger on the curve does as well as a formula for the first reading.
Cold and draughts. Sensitive pupils move away from the window before the airing.

If things do not go to plan

The box has only the SCD41. The lesson works; temperature and humidity come from the SCD41 and the VOC column stays empty.
The window cannot be opened. Air with the door to the corridor only, and say so: the fall will be slower, which is itself a result.
The CO_2 barely rises. A small class in a large or leaky room. Read the rate per pupil, or repeat with the full class.
The laptop went to sleep. The rows recorded until then are in the CSV in `MagicBox\exports`; start a new recording and put the two files together in Excel.
The box will not connect. Use the illustrative example in "What the data should look like" as a dataset and draw the curve on the board.

7. Further resources

- **German Federal Environment Agency (UBA)** — `https://www.umweltbundesamt.de` — search for the guidance on CO_2 and airing in schools (*Lüften in Schulen*), with the 1000 and 2000 ppm reference values.
- **NOAA Global Monitoring Laboratory** — `https://gml.noaa.gov/ccgg/trends/` — the outdoor CO_2 measured month by month, the floor of every classroom curve.
- **REHVA** — `https://www.rehva.eu` — classroom ventilation and the standard EN 16798-1.
- **Sensirion** — `https://sensirion.com` — how the SCD41 measures CO_2 and what the SGP41 VOC index means.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *a five-minute break: do you open the door or the window? Why?*

For a more formal judgement, use the **common three-axis rubric**:

- **Measurement practice** — the pupil starts the recording in an aired room, marks every event on time and keeps the box away from the window and from people.
- **Data interpretation** — the pupil links each bend of the curve to an event and computes the rate of rise and the halving time.
- **Scientific reasoning** — the pupil explains why the CO_2 cannot fall below the outdoor level and why the door and the window differ, and states a limitation of the measurement.

The rubric itself, with descriptors for all levels, is defined in the *curriculum plan*.

9. Pupil sheet

The sheet is a separate document, in the folder `docs/en/student_sheets`: the file `breathing_in_the_classroom.pdf`. It fits on one sheet, printed on both sides, built on the three layers: what I see on the screen, what I do on my own, the conclusion I write. Its event log is kept by the recorder during the lesson; the questions on the back are for lesson 2, with the Excel file open.